Conduction classes, energy bands & forbidden gaps

30-order resistivity scale, opposite temperature laws, filled-band theorem, and key gap values.

30-order resistivity scale & classification

Electrical resistivity spans ~30 orders of magnitude at 300 K: metals sit near 10^(-8) ohm.m, semiconductors span 10^(-5) to 10^4 ohm.m, and insulators reach 10^(18) ohm.m.

$$\rho_{\mathrm{Cu}} \approx 1.7 \times 10^{-8}\ \Omega\cdot\mathrm{m}, \quad \rho_{i,\,\mathrm{Si}} \approx 2300\ \Omega\cdot\mathrm{m}$$

$$\rho_{\mathrm{quartz}} \sim 10^{17}\ \text{to}\ 10^{19}\ \Omega\cdot\mathrm{m}$$

WHY / Magnitude alone overlaps at boundaries — what truly separates classes physically is whether carrier count n is FIXED (metal) or THERMALLY ACTIVATED across a gap (semiconductor)!

WATCH / Mystery sample E1 with rho ~ 10^(-2) ohm.m whose resistance FALLS when heated is an extrinsic/semiconducting sample, NEVER a nichrome alloy!

Temperature coefficient alpha: metal vs Si vs NTC

Conductivity sigma = n e mu pulls two temperature levers in opposite directions: lattice phonon scattering lowers mu(T), whereas thermal excitation raises n(T).

$$\text{Metal}: \ n = \text{const}, \ \mu\downarrow \Rightarrow \frac{d\rho}{dT} > 0 \ (\alpha > 0)$$

$$\text{Semiconductor / NTC}: \ n_i(T)\uparrow\uparrow \Rightarrow \frac{d\rho}{dT} < 0 \ (\alpha < 0)$$

WHY / In silicon, mild acoustic scattering mu ~ T^(-3/2) is completely crushed by exponential bond breaking exp(-Eg / 2kT), plunging resistivity steeply on warming!

WATCH / NTC thermistors exploit steep negative dR/dT for precision temperature sensing and inrush-current limiting; never treat a thermistor as ohmic over wide T!

Allowed bands & why a filled band CANNOT conduct

Bringing N isolated atoms into a periodic crystal splits discrete atomic orbitals into N closely spaced crystal states forming valence (VB) and conduction (CB) bands.

$$\sum_{\text{full band}} \mathbf{v}(\mathbf{k}) = \mathbf{0} \quad (\text{every } +\mathbf{k} \text{ paired with } -\mathbf{k})$$

WHY / Electric force F = -eE must shift electrons into adjacent empty k-states! A 100% full valence band has ZERO empty states — like a bumper-to-bumper parking lot, net current is identically zero!

WATCH / Never answer "electrons are too tightly bound to move" on JEE conceptual questions — answer "no unoccupied quantum states exist inside a filled band"!

Forbidden band-gap benchmarks at 300 K

Forbidden energy gap Eg = Ec - Ev decides room-temperature carrier population relative to thermal energy kT = 25.9 meV at 300 K:

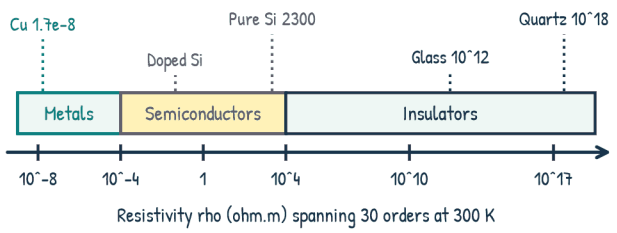
$$E_g: \ \mathrm{Ge} = 0.66\ \mathrm{eV}, \quad \mathrm{Si} = 1.12\ \mathrm{eV}, \quad \mathrm{GaAs} = 1.42\ \mathrm{eV}$$

$$\mathrm{C\ (diamond)} = 5.5\ \mathrm{eV} \quad (\text{insulator}: \ E_g \gg k_B T!)$$

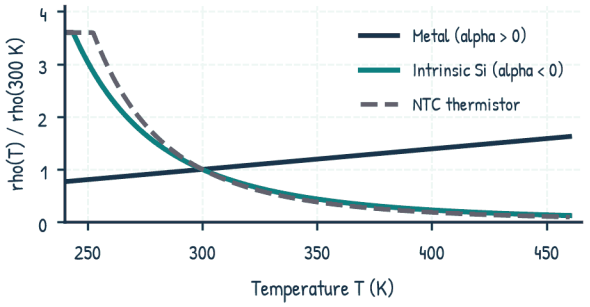
WHY / At 300 K, ni(C)/ni(Si) ~ exp[-(5.5 - 1.12)/(2 x 0.0259)] ~ 10^(-37) (essentially zero free electrons in diamond!), while ni(Ge)/ni(Si) ~ exp(+8.8) ~ 7 x 10^3!

WATCH / Carbon and silicon share Group-14 valence 4 and diamond tetrahedral bonding — ONLY smaller atomic spacing makes C 5.5 eV insulating vs Si 1.12 eV semiconducting!

30-order logarithmic resistivity axis from copper to fused quartz



Opposite temperature laws: scattering in metals vs pair creation in Si



REMEMBER / Metals warm UP in resistance (phonon scattering with fixed n); semiconductors cool DOWN in resistance (exponential carrier creation beats mobility loss)!

Intrinsic carriers, doping, neutrality & mass action

Half-gap thermal law, bond-swapping holes, Group-15/13 impurities, and np = ni^2 seesaw.

Intrinsic carrier law & exact half-gap exponent

In pure crystal, thermal vibration breaks covalent bonds in pairs, so electron and hole concentrations are strictly equal: n = p = ni.

$$n_i(T) = \sqrt{N_c N_v}\, e^{-E_g/(2k_BT)} \propto T^{3/2} e^{-E_g/(2k_BT)}$$

$$\text{Si (300 K)}: \quad n_i = 1.5 \times 10^{16}\ \text{m}^{-3}, \quad \frac{E_g}{2k_BT} = 21.6$$

WHY / Notice the factor 2 in exp(-Eg / 2kT)! Doubling T from 300 K to 600 K boosts exp(...) by exp(10.8) = 5.0e4 and T^(3/2) by 2.83 => total 1.43e5x jump!

WATCH / Audit check (Fig F27.5): for Si from 300 K to 400 K, exact ni(400)/ni(300) = (4/3)^(3/2) exp(5.41) = 346 (NOT 800)! Always keep 2kT in the denominator!

Holes as positive quasiparticles & Group-15/13 doping

When a broken bond leaves an empty valence state, neighbouring valence electrons hop left under E right — making the vacancy drift right exactly like charge +e!

$$\text{Group 15 (P, As, Sb)} \rightarrow \text{n type } (+e \text{ ion})$$

$$\text{Group 13 (B, Al, Ga)} \rightarrow \text{p type } (-e \text{ ion})$$

WHY / Audit note: Si diamond lattice has N_Si = 8 / (0.543 nm)^3 = 5.0 x 10^28 atoms/m^3, so Nd = 10^21 m^(-3) is 1 P per 5 x 10^7 Si atoms (1 ppm is 5 x 10^22 m^(-3))!

WATCH / Ionised donor cores P^+ and acceptor cores B^- are LOCKED in the crystal lattice — they set space charge rho(x) but carry ZERO conduction current!

Bulk electrical neutrality & exact quadratic law

Every donated electron came from an initially neutral dopant atom, so an isolated n-type or p-type crystal has ZERO net charge!

$$n + N_a^- = p + N_d^+, \qquad n\,p = n_i^2$$

$$n = \frac{(N_d - N_a) + \sqrt{(N_d - N_a)^2 + 4n_i^2}}{2} \quad \text{(exact!)}$$

WHY / Whenever |Nd - Na| >> ni (standard extrinsic regime at 300 K), square root expands cleanly to majority n ~ Nd - Na while minority collapses to p = ni^2 / n!

WATCH / Classic JEE Advanced trap: if BOTH donors Nd and acceptors Na are added ("compensated Si"), net majority type is set by SIGN of (Nd - Na), and Nd = Na restores n = p = ni!

Ten-order majority/minority seesaw (E3 & Q5)

Adding donors floods the crystal with electrons, increasing electron-hole recombination until minority holes drop inversely so the product np stays locked at ni^2:

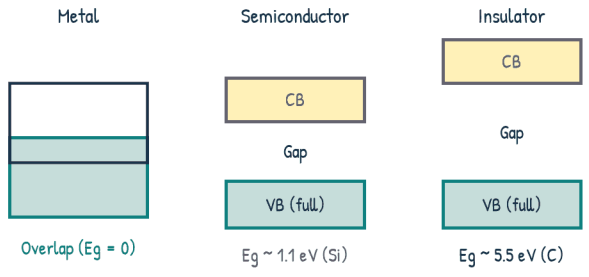
$$N_d = 10^{21}\ \text{m}^{-3} \Rightarrow n \approx 10^{21}\ \text{m}^{-3}, \quad p = \frac{n_i^2}{N_d} = 2.25 \times 10^{11}\ \text{m}^{-3}$$

$$\frac{n}{p} = \left(\frac{N_d}{n_i}\right)^2 \approx 4.4 \times 10^9 \quad \text{(10 orders apart!)}$$

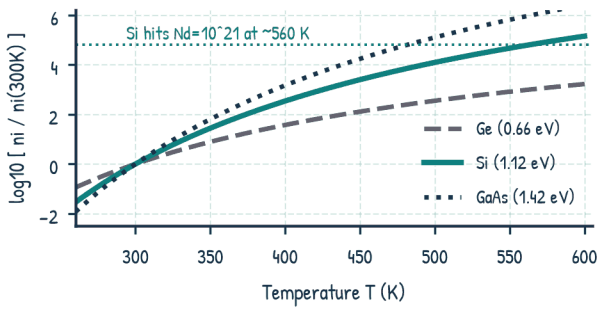
WHY / Dynamic equilibrium generation rate G(T) depends ONLY on temperature T, whereas recombination R ~ n.p: setting R = G forces n.p = ni(T)^2 even after heavy doping!

WATCH / Always convert cm^(-3) <-> m^(-3) cleanly BEFORE squaring ni: 1 cm^(-3) = 10^6 m^(-3), so ni(Si) = 1.5 x 10^10 cm^(-3) = 1.5 x 10^16 m^(-3)!

Allowed valence/conduction bands and forbidden gap across solids



Exponential thermal growth of intrinsic carrier density ni(T)



REMEMBER / Doped crystals are ELECTRICALLY NEUTRAL overall; n.p = ni^2 holds in thermal equilibrium for BOTH pure and doped semiconductors!

Drift, mobility, conductivity & Fick diffusion

Dual-carrier drift currents, why e- and h+ currents ADD, and Einstein relation D/mu = kT/e.

Drift speed, mobility asymmetry & total conductivity

Under weak electric field E, electrons drift opposite E with v_e = mu_e E while positive holes drift along E with v_h = mu_h E — BOTH conventional currents point ALONG E!

$$J_{\text{drift}} = e(n\mu_e + p\mu_h)E \Rightarrow \sigma = \tfrac{1}{\rho} = e(n\mu_e + p\mu_h)$$

$$\text{Si (300 K)}: \quad \mu_e = 0.135,\ \mu_h = 0.048\ \text{m}^2/(\text{V}\cdot\text{s}) \quad (\tfrac{\mu_e}{\mu_h} = 2.81)$$

WHY / Conduction-band electrons have lighter curvature effective mass and looser wavefunctions than valence-band bond-hopping holes, making NPN & NMOS faster than PNP & PMOS!

WATCH / Never subtract n mu_e - p mu_h in drift conductivity! (-e)(-v_e) + (+e)(+v_h) = +e(n v_e + p v_h) — opposite charges moving in opposite directions ADD!

Intrinsic vs doped silicon numbers (E2, E3 & P21)

Pure silicon conducts like wet stone; trace Group-15/13 doping boosts conductivity by 5 orders of magnitude without altering crystal geometry:

$$\sigma_i = 1.6\times10^{-19}(1.5\times10^{16})(0.183) = 4.39\times10^{-4}\ \text{S/m}$$

$$\rho_i = 2280\ \Omega\cdot\text{m}; \quad N_d = 10^{21} \Rightarrow \sigma_n \approx eN_d\mu_e = 21.6\ \text{S/m}$$

WHY / Minority hole contribution e p mu_h ~ 1.7e-9 S/m is 10 orders smaller than 21.6 S/m — drop the minority term instantly whenever doping exceeds 100 ni!

WATCH / Minimum conductivity of a doped semiconductor does NOT occur at n = p = ni! Minimising n mu_e + (ni^2/n) mu_h gives p-type minimum at p = ni sqrt(mu_e / mu_h)!

Fick diffusion current & sign rule

Even when E = 0, random thermal motion drives net particle flux from high to low concentration (-D dn/dx), creating diffusion current density:

$$J_{n,\,\text{diff}} = (-e)\left(-D_n\tfrac{dn}{dx}\right) = +eD_n\tfrac{dn}{dx}$$

$$J_{p,\,\text{diff}} = (+e)\left(-D_p\tfrac{dp}{dx}\right) = -eD_p\tfrac{dp}{dx}$$

WHY / If BOTH electrons and holes decrease toward +x (dn/dx < 0, dp/dx < 0, e.g. surface illumination), both diffuse right — so J_p points +x while J_n points -x!

WATCH / Watch the PLUS sign on J_{n,diff} = +e Dn (dn/dx) vs MINUS sign on J_{p,diff} = -e Dp (dp/dx) — opposite of drift because same-direction particle fluxes carry opposite charges!

Total transport equations & Einstein relation

Combining drift and diffusion for each carrier species and enforcing zero net current in thermal equilibrium links diffusion constant D directly to mobility mu:

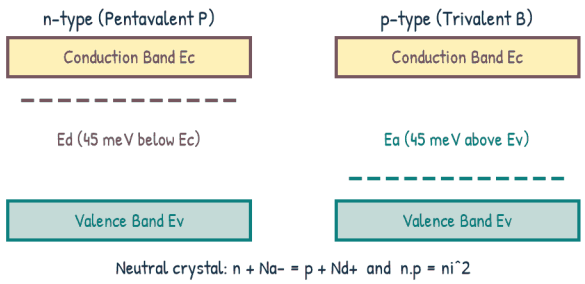
$$J_n = en\mu_e E + eD_n\tfrac{dn}{dx}, \qquad J_p = ep\mu_h E - eD_p\tfrac{dp}{dx}$$

$$\tfrac{D_n}{\mu_e} = \tfrac{D_p}{\mu_h} = \tfrac{k_BT}{e} = V_T = 25.9\ \text{mV} \quad (300\ \text{K})$$

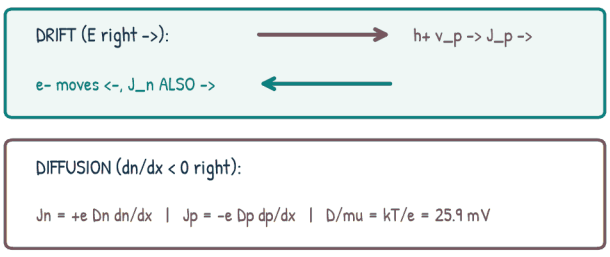
WHY / Both scattering relaxation time tau and thermal energy kT govern random walks AND field drift — their ratio D/mu depends ONLY on temperature T!

WATCH / In Si at 300 K: Dn = 0.135 x 0.0259 = 3.50 x 10^(-3) m^2/s (35 cm^2/s) and Dp = 1.24 x 10^(-3) m^2/s (12.4 cm^2/s) — quick sanity numbers for Olympiad checks!

Shallow donor Ed and acceptor Ea levels inside the forbidden gap



Drift vs concentration-gradient diffusion currents and Einstein relation



REMEMBER / Under electric field E, electron and hole drift currents ADD (sigma = e(n mu_e + p mu_h)); under a concentration gradient, their diffusion currents SUBTRACT!

p-n junction equilibrium, barrier hill & space charge

Uncovered ion dipole, Boltzmann contact potential V0, and aligned rho(x), E(x), V(x) curves.

Depletion formation & drift-diffusion cancellation

Joining p and n silicon triggers massive majority diffusion across x = 0, uncovering immobile negative acceptors (-e Na) on p-side and positive donors (+e Nd) on n-side.

$$J_{n,\text{drift}} + J_{n,\text{diff}} = 0, \qquad J_{p,\text{drift}} + J_{p,\text{diff}} = 0$$

$$N_a\, x_p = N_d\, x_n \quad (\text{total space charge } Q_+ + Q_- = 0!)$$

WHY / Overall dipole neutrality Na xp = Nd xn forces the depletion region to penetrate DEEPER into the LIGHTLY doped side (if Na >> Nd, almost all W sits inside n)!

WATCH / Trap 2: "The depletion region has net positive/negative charge" -> FALSE! Equal +Q and -Q layers cancel identically!

Built-in contact potential V0 from Boltzmann

Integrating e n mu_e (-dV/dx) + e Dn (dn/dx) = 0 from neutral p-bulk to neutral n-bulk yields the exact built-in barrier height V0:

$$V_0 = V_T \ln\!\left(\frac{N_a N_d}{n_i^2}\right) = \frac{k_B T}{e} \ln\!\left(\frac{n_{n0}}{n_{p0}}\right)$$

$$N_a = N_d = 10^{22}\ \text{m}^{-3} \Rightarrow V_0 = 0.695\ \text{V}$$

WHY / Typical built-in barriers sit just below Eg/e: ~0.65-0.75 V in Si, ~0.25-0.35 V in Ge, and ~1.1-1.2 V in GaAs!

WATCH / Why can you NOT power a bulb with V0 = 0.7 V? Metal wire contacts at both ends create equal and opposite contact potentials summing to ZERO around any closed loop!

Exact aligned rho(x), E(x) & V(x) profiles

Integrating Poisson dE/dx = rho / eps twice across [-xp, +xn] connects rectangular charge blocks to triangular field and parabolic S-curve potential:

$$|E_{\max}| = \frac{eN_a x_p}{\varepsilon} = \frac{eN_d x_n}{\varepsilon} = \frac{2V_0}{W}$$

$$\text{Area under } -E(x) = \tfrac{1}{2}|E_{\max}|(x_p + x_n) = V_0$$

WHY / Peak electric field |E_max| sits RIGHT AT THE METALLURGICAL JUNCTION x = 0 (~5 x 10^5 V/m across 2.5 um!), and drops linearly to zero at edges -xp and +xn!

WATCH / Electron potential energy U_e(x) = -e V(x) is UPSIDE DOWN relative to electrostatic V(x): V(x) rises p -> n, so band edges Ec(x), Ev(x) step UP n -> p!

Depletion width scaling W ~ sqrt(V0 + VR)

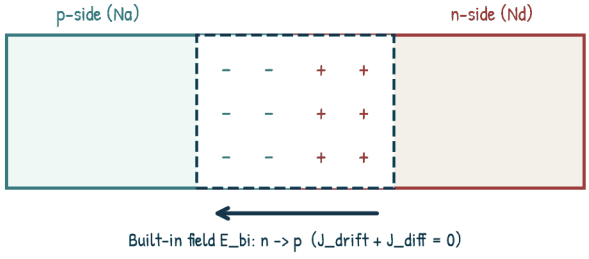
Applying external voltage V shifts the potential barrier to (V0 - V) in forward bias or (V0 + VR) under reverse voltage VR:

$$W(V_R) = \sqrt{\frac{2\varepsilon(V_0 + V_R)}{e\, N_{\text{eff}}}}, \qquad \frac{1}{N_{\text{eff}}} = \frac{1}{N_a} + \frac{1}{N_d}$$

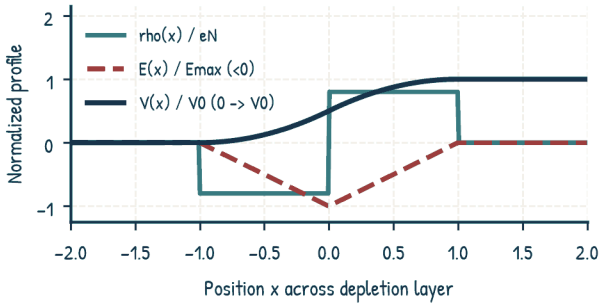
WHY / Forward bias shrinks W and lowers the hill for exponential majority injection; reverse bias widens W and raises the hill to V0 + VR!

WATCH / Q9 & P23 check: quadrupling VR from 1 V to 4 V doubles W ONLY when VR >> V0! With V0 = 0.7 V, exact ratio is sqrt((0.7+4)/(0.7+1)) = 1.66x!

Depletion space-charge dipole and opposing drift-diffusion balance



Aligned space charge rho(x), electric field E(x), and barrier hill V(x)



REMEMBER / Built-in E_bi points from n (+Nd ions) to p (-Na ions), so n-side sits HIGHER in electrostatic potential by V0 — yet a voltmeter across the terminals ALWAYS reads ZERO!

Shockley diode law, 60 mV rule, dynamic rd & breakdown

Exponential forward conduction, decade ladder, static vs dynamic resistance, Zener vs avalanche.

Shockley equation & the 60 mV-per-decade ladder

Lowering the junction hill from V0 to V0 - V boosts majority carrier injection over the barrier by exp(eV / kT), yielding Shockley's diode equation:

$$I = I_0(e^{V/(\eta V_T)} - 1), \qquad V_T = \frac{k_B T}{e} = 25.9 \text{ mV } (300 \text{ K})$$

$$\Delta V_{10\times} = \eta V_T \ln 10 = 59.5\,\eta \text{ mV/dec } (\approx 60 \text{ mV})$$

WHY / Example E5 & Q10: 1 mA at 0.60 V -> 10 mA at 0.66 V -> 100 mA at 0.72 V -> 1 A at 0.78 V! The "0.7 V knee" is simply where nanoamps cross into milliamps!

WATCH / Real discrete Si diodes often have recombination factor eta ~ 1.5-2 at low current (90-120 mV/decade), whereas BJT collector current IC(VBE) obeys pure eta = 1!

Static DC resistance R_DC vs AC dynamic resistance r_d

JEE Main & NEET frequently test small voltage ripples superimposed on a DC forward diode bias — NEVER divide V / I when they ask for dynamic resistance!

$$R_{DC} = \frac{V}{I}, \qquad r_d = \left(\frac{dI}{dV}\right)^{-1} = \frac{\eta V_T}{I + I_0} \approx \frac{26 \text{ mV}}{I \text{ (mA)}} \ \Omega$$

WHY / At I = 2 mA and V = 0.70 V, static R_DC = 350 ohm, whereas AC slope resistance is rd = 26 mV / 2 mA = 13 ohm — 27 times smaller!

WATCH / Near V = 0 (|V| << VT), exp(V/VT) - 1 ~ V/VT becomes linear and ohmic with zero-bias slope resistance r0 = VT / I0!

Reverse saturation I0 & its 10-degree doubling rule

For reverse bias V < -4 VT (~ -0.1 V), exp(V/VT) vanishes and reverse current saturates at -I0 (nA in Si, uA in Ge), independent of reverse voltage before breakdown.

$$I \approx -I_0 \propto n_i^2 \propto T^3 e^{-E_g/(k_B T)}$$

WHY / Notice NO factor of 2 in exp(-Eg / kT) because minority concentration p_{n0} = ni^2 / Nd scales as ni SQUARED! I0 roughly DOUBLES every +7 to +10 K rise!

WATCH / P27 subtlety: at fixed V = 0.60 V, heating shrinks exp(V/VT) yet TOTAL diode current STILL RISES on heating because I0 ~ exp(-Eg/kT) explodes faster, dropping V_knee by ~2 mV/K!

Zener tunnelling (<5 V) vs avalanche breakdown (>6 V)

Two distinct physical mechanisms trigger sharp reverse current rise at breakdown voltage V_BR:

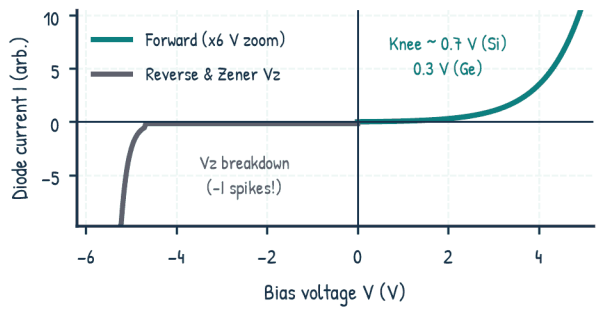
$$V_Z < 5 \text{ V } (W < 10 \text{ nm}): \text{ Zener quantum tunnelling}$$

$$V_{BR} > 6 \text{ V } (\text{wider } W): \text{ avalanche impact ionisation}$$

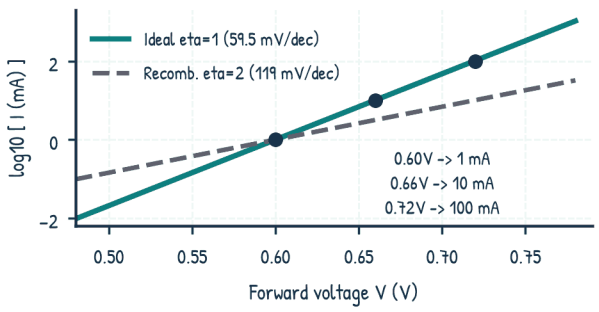
WHY / Warming shrinks Eg so Zener tunnelling gets EASIER (dVz/dT < 0), whereas warming shortens mean free path so avalanche needs MORE voltage (dV_BR/dT > 0)!

WATCH / Around Vz = 5.1-5.6 V both effects cancel to give dVz/dT ~ 0 — why 5.1 V and 5.6 V Zeners are precision industry favourites (P17)!

Full junction diode I-V characteristic from Zener breakdown to knee



Room-temperature semilog ladder: 59.5 mV per decade of forward current



REMEMBER / Every +60 mV forward bias multiplies ideal diode current by 10x at 300 K; small-signal dynamic resistance is rd = VT / I = 26 mV / I_DC!

Half/full-wave rectifiers, capacitor ripple & clippers

Exact DC, RMS, efficiency, form factor, PIV, bridge double drop, filter ripple, and clippers/clampers.

Half-wave vs full-wave rectifier benchmark table

For sinusoidal input v(t) = Vp sin(wt) feeding pure load RL (ideal diodes):

$$\text{HW}: \quad V_{\text{dc}} = \frac{V_p}{\pi}, \quad V_{\text{rms}} = \frac{V_p}{2}, \quad \eta = 40.6\%$$

$$\text{FW}: \quad V_{\text{dc}} = \frac{2V_p}{\pi}, \quad V_{\text{rms}} = \frac{V_p}{\sqrt{2}}, \quad \eta = 81.2\%$$

WHY / AC ripple factor r = sqrt((Vrms/Vdc)^2 - 1) drops from 1.21 (half-wave: more AC than DC!) down to 0.482 (full-wave), and fundamental ripple frequency DOUBLES from f to 2f!

WATCH / PIV (Peak Inverse Voltage) trap: half-wave = Vp; 4-diode bridge = Vp; 2-diode CENTRE-TAP full-wave = 2 Vp across the blocking diode!

Real silicon diode drops: 4-diode bridge vs centre-tap

When transformer peak Vp is modest (e.g. 5-12 V), always check whether the question specifies silicon diodes (V_gamma = 0.7 V) or ideal diodes:

$$\text{Bridge (4 diodes)}: \quad V_{p,\text{out}} = V_p - 2V_\gamma \approx V_p - 1.4 \text{ V}$$

$$\text{Centre-tap (2 diodes)}: \quad V_{p,\text{out}} = V_p - V_\gamma \approx V_p - 0.7 \text{ V}$$

WHY / Trace the diamond loop: during positive half-cycle D1 -> RL -> D3 conduct IN SERIES; during negative half-cycle D2 -> RL -> D4 conduct IN SERIES!

WATCH / If ONE diode in a 4-diode bridge burns OPEN, it degrades cleanly into a HALF-WAVE rectifier (f_rip = f, Vdc halves); if one shorts, it shorts the AC transformer!

Shunt capacitor filter & sawtooth ripple equation

Placing large capacitor C in parallel with RL charges C to peak Vp every cycle, then discharges nearly linearly at rate dV/dt = -I_L / C for time T_dis:

$$\Delta V_{\text{FW}} \approx \frac{I_L}{2fC} = \frac{V_p}{2fR_LC}, \qquad \Delta V_{\text{HW}} \approx \frac{I_L}{fC}$$

$$V_{\text{dc, filtered}} \approx V_p - \frac{\Delta V}{2}$$

WHY / E7 & P16 check: full-wave 50 Hz, I = 100 mA, C = 1000 uF gives dV = 0.1 / (100 x 10^-3) = 1.0 V; 50 mA with 470 uF gives dV = 1.06 V!

WATCH / Match 2f vs f CAREFULLY! Full-wave recharges C twice per mains period (T/2 = 10 ms at 50 Hz), half-wave only once (T = 20 ms -> twice the ripple)!

Clippers (limiters), DC clampers & peak detectors

Diodes reshape AC waveforms either by slicing above/below a threshold (clipper: resistor + diode) or shifting DC baseline without distortion (clamper: capacitor + diode):

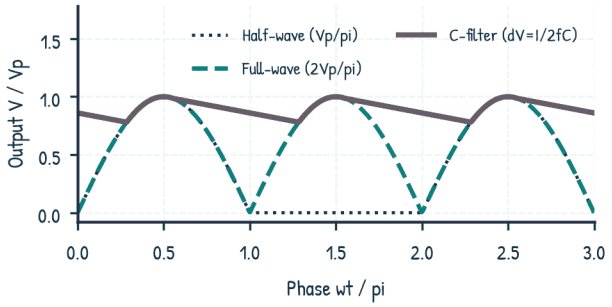
$$\text{Biased shunt clipper}: \quad v_{\text{out}} = \min(v_{\text{in}},\ V_{\text{ref}} + V_\gamma)$$

$$\text{Pos. clamper}: \quad v_{\text{out}}(t) = v_{\text{in}}(t) + V_p \geq 0$$

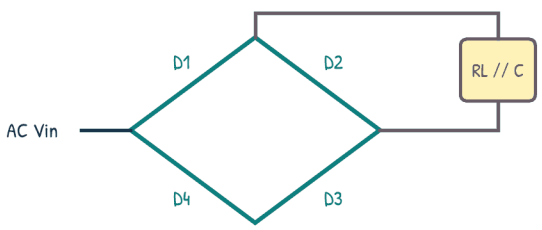
WHY / Always solve ANY clamper in two steps: (1) find the half-cycle that turns the diode ON first to lock capacitor voltage V_C; (2) write KVL v_out = v_in +/- V_C on the OFF half-cycle!

WATCH / A clamper NEVER alters peak-to-peak swing 2 Vp of the input wave — it translates the entire waveform vertically from [-Vp, +Vp] to [0, +2 Vp] (or [-2 Vp, 0])!

Half-wave, full-wave, and capacitor-filtered sawtooth ripple waveforms



Four-diode full-wave bridge rectifier path and double-drop arithmetic



2 series diodes ON each half-cycle => Vp_out = Vp - 2*0.7V; PIV = Vp

REMEMBER / Full-wave bridge drops TWO knees (Vp - 1.4 V) with PIV = Vp and ripple f_rip = 2f; smoothing ripple is dV = I / (2fC) full-wave vs I / (fC) half-wave!

Zener regulators, LEDs, photodiodes & solar cells

Two-sided Zener resistor squeeze, direct vs indirect bandgaps, and 4-quadrant PV physics.

Zener shunt regulator & worst-case two-sided squeeze

Reverse-biased Zener diode clamps load voltage at V_out = Vz PROVIDED Zener current stays between knee minimum IZ,min and power limit IZ,max = PZ,max / Vz:

$$I_R = \frac{V_{\text{in}} - V_Z}{R_s} = I_Z + I_L, \qquad R_{s,\,\max} = \frac{V_{\text{in,\,min}} - V_Z}{I_{L,\,\max} + I_{Z,\,\min}}$$

$$I_{Z,\,\max} = \frac{V_{\text{in,\,max}} - V_Z}{R_s} - I_{L,\,\min} \quad (\text{worst case at } I_L = 0!)$$

WHY / E8 & Q16: Vin = 12 V, Vz = 5 V, IL = 0..50 mA, IZ,min = 5 mA => Rs <= 7 / 0.055 = 127 ohm; picking 120 ohm gives no-load IZ = 58.3 mA, PZ = 0.29 W!

WATCH / Before writing V_out = Vz in ANY JEE problem, ALWAYS check Thevenin open-circuit voltage V_th = Vin RL / (Rs + RL) >= Vz! If V_th < Vz, the Zener is OFF and V_out = V_th!

Light-Emitting Diodes (LEDs) & why silicon fails

Forward-biasing a heavily doped direct-bandgap junction injects electrons & holes that recombine radiatively across Eg, emitting monochromatic photons:

$$\lambda_{\text{peak}} = \frac{hc}{E_g} \approx \frac{1240\ \text{eV}\cdot\text{nm}}{E_g\ (\text{eV})}, \qquad E_g \geq 1.8\ \text{eV for visible}$$

WHY / Visible anchors: Red 620-680 nm (1.8-2.0 eV, GaAsP), Green 550 nm (2.25 eV, GaP), Blue 470 nm (2.64 eV, InGaN); IR remotes use GaAs (1.42 eV -> 873 nm)!

WATCH / Why NO silicon LEDs (E9 & Q18)? Two fatal reasons: (1) Eg = 1.12 eV -> 1107 nm infrared, AND (2) Si has an INDIRECT gap (momentum mismatch goes into lattice heat phonons)!

Photodiodes (Quadrant III reverse-bias light meters)

Illuminating a reverse-biased p-n junction with photons hv > Eg generates electron-hole pairs inside/near the depletion region that E_bi sweeps apart instantly:

$$I_{\text{total}} = -(I_0 + I_{\text{photo}}), \qquad I_{\text{photo}} = \eta_{\text{QE}}\, e\, \frac{P_{\text{opt}}}{h\nu} = \mathcal{R}_\lambda P_{\text{opt}}$$

WHY / Why reverse bias instead of forward? Reverse bias (1) crushes dark current I0 to nanoamps so tiny light shifts stand out, AND (2) widens W -> shrinks Cj for nanosecond speed!

WATCH / Photons with lambda > 1240 / Eg pass straight through the crystal unabsorbed — a silicon photodiode is completely blind beyond ~1100 nm!

Solar photovoltaic cells (Quadrant IV zero-bias power)

With NO external battery, photogenerated carriers swept across the junction pile holes on p and electrons on n, forward-biasing the cell to deliver real electrical power:

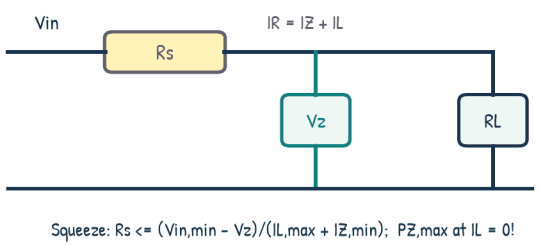
$$I(V) = I_0\left(e^{\,V/V_T} - 1\right) - I_L, \qquad V_{\text{oc}} = V_T \ln\!\left(1 + \frac{I_L}{I_0}\right)$$

$$\eta_{\text{PV}} = \frac{P_{\text{max}}}{P_{\text{in}}} = \frac{\text{FF}\cdot V_{\text{oc}}\, I_{\text{sc}}}{P_{\text{in}}} \quad (\text{Quadrant IV}:\ V>0,\ I<0)$$

WHY / Choose active layer with Eg ~ 1.1-1.5 eV (Si 1.12 eV, GaAs 1.42 eV, CdTe 1.45 eV) and high optical absorption alpha so maximum solar spectrum converts into power!

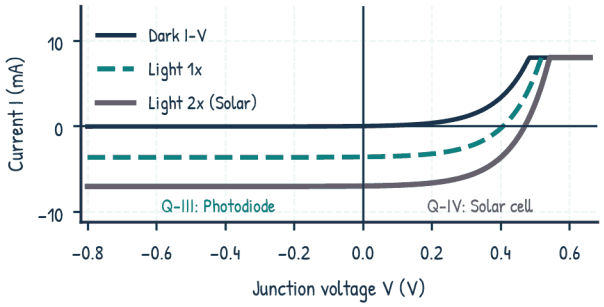
WATCH / Q20 favourite JEE question: "A solar cell operates in Quadrant IV because it generates power (V > 0 while current exits the positive p-terminal)" — NEVER draw external bias!

Zener shunt voltage regulator and two-sided worst-case resistor squeeze



Squeeze: Rs <= (Vin,min - Vz)/(IL,max + IZ,min); PZ,max at IL = 0!

Illuminated p-n junction across Quadrant III (sensor) and IV (PV cell)



REMEMBER / Size Rs at (Vin,min & IL,max), then check Zener power PZ,max at (Vin,max & IL = 0 disconnected load!); solar cells operate in Quadrant IV (V > 0, I < 0)!

Bipolar junction transistor (BJT): gains & modes

Asymmetric doping geometry, alpha-beta identities, saturation test, and mode table.

Asymmetric E-B-C design & current law

An NPN transistor is NOT two back-to-back discrete diodes: Emitter is heavily doped (n++), Base is ultra-thin & lightly doped (p), and Collector has the largest area (n):

$$I_E = I_C + I_B, \qquad \alpha = \frac{I_C}{I_E}\ (<1), \qquad \beta = \frac{I_C}{I_B} = \frac{\alpha}{1-\alpha}$$

$$\alpha = \frac{\beta}{\beta+1}, \qquad I_E = (\beta+1) I_B \quad (\text{active mode ONLY!})$$

WHY / Thin lightly doped base lets 98-99.5% of injected emitter electrons diffuse across to the collector before recombining (alpha = 0.99 => beta = 99, E10 & P10)!

WATCH / Collector has the LARGEST physical area NOT for doping, but to dissipate heat P_C = V_CE I_C (~99% of transistor heating occurs at the reverse-biased B-C junction)!

Four operating modes set by E-B & C-B junction biases

Every BJT question starts by classifying the bias polarities of Emitter-Base (EBJ) and Collector-Base (CBJ) junctions:

$$\text{Cutoff}: \ I_C = 0 \quad | \quad \text{Active}: \ I_C = \beta I_B$$

$$\text{Sat}: \ V_{CE,\,\text{sat}} \approx 0.2\ \text{V}, \quad I_C = I_{C,\,\text{sat}} < \beta I_B$$

WHY / Reverse-active (EBJ OFF, CBJ ON) swaps emitter and collector roles with terrible gain beta_rev ~ 0.1-2 because the collector was never heavily doped to inject!

WATCH / Swapping Emitter and Collector pins of an NPN transistor on a breadboard puts it into reverse-active mode — circuit barely works instead of burning out!

20-second saturation test (Q23 & P25 killer trap!)

Given VCC, collector resistor RC, base current IB, and gain beta, NEVER write IC = beta IB blindly — run this 3-step check first:

1. $I_{C,\,\text{trial}} = \beta I_B$
2. $I_{C,\,\text{max}} = \frac{V_{CC} - V_{CE,\,\text{sat}}}{R_C}$
3. $I_C = \min(\beta I_B,\ I_{C,\,\text{max}})$ $\quad (\beta I_B \geq I_{C,\,\text{max}} \Rightarrow \text{SAT!})$

WHY / Q23 proof: VCC = 5 V, RC = 1 k, beta = 100, IB = 100 uA => beta*IB = 10 mA gives impossible VCE = 5 - 10 = -5 V! True saturated IC = (5 - 0.2)/1k = 4.8 mA!

WATCH / Marking-scheme penalty warning (Part 12): writing IC = beta IB in saturation loses ALL numerical marks even with flawless arithmetic!

DC load line & Q-point placement

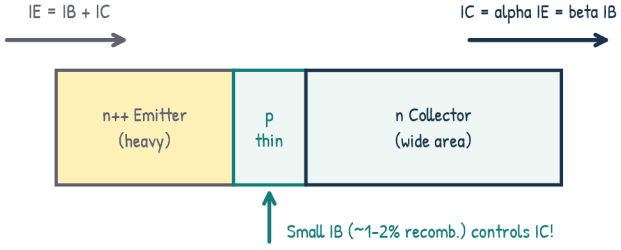
Kirchhoff's voltage loop on the collector side imposes a straight line across the transistor's output characteristic family:

$$V_{CE} = V_{CC} - I_C(R_C + R_E) \Rightarrow I_C = -\frac{V_{CE}}{R_C + R_E} + \frac{V_{CC}}{R_C + R_E}$$

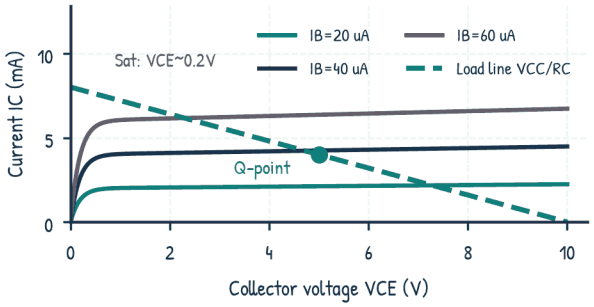
WHY / Placing quiescent operating point Q near the centre (VCE,Q ~ VCC / 2) maximises symmetric peak-to-peak undistorted output swing before hitting cutoff or saturation clipping!

WATCH / Where does amplified signal power come from (C12 & Trap 8)? 100% from the DC battery VCC — the base signal merely modulates the valve opening!

Asymmetric doping and thin-base electron transport in an NPN transistor



Collector family IC(VCE), DC load line, active Q-point, and saturation wall



REMEMBER / IC = beta IB holds ONLY in forward-active mode! ALWAYS test beta*IB against (VCC - 0.2 V)/RC FIRST — whichever is SMALLER wins!

Common-emitter amplifier, gains & BJT switch design

Small-signal gm, voltage/power gain, 180-deg phase inversion, and hard-saturated load switches.

Common-emitter (CE) small-signal gains & transconductance

Around active Q-point (IC, VCE), tiny input wiggle dV_BE produces dI_B = dV_BE / r_in and collector wiggle dI_C = beta dI_B = gm dV_BE:

$$g_m = \frac{\Delta I_C}{\Delta V_{BE}} = \frac{I_C}{V_T}, \qquad r_\pi = \frac{\Delta V_{BE}}{\Delta I_B} = \frac{\beta}{g_m} = \frac{\beta V_T}{I_C}$$

$$A_v = \frac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} = -\beta\frac{R_C}{R_{\text{in}}} = -g_m R_C, \qquad A_p = \beta\,|A_v| = \beta^2\frac{R_C}{R_{\text{in}}}$$

WHY / MINUS sign in Av = -gm RC means 180-deg phase inversion (Q24): rising Vin drives higher IC, larger I_C R_C drop, and LOWER output voltage VCE = VCC - IC RC!

WATCH / Notice power gain Ap = A_i |A_v| scales as BETA SQUARED! For beta = 100 and RC = R_in, |Av| = 100 while signal power gain is Ap = 10,000 (+40 dB)!

CB vs CE vs CC configuration comparison

Only Common-Emitter gives BOTH high current gain (beta) and high voltage gain (gm RC), making CE the universal workhorse stage:

$$\text{CE}:\ A_i = \beta,\ |A_v| \gg 1,\ \Delta\phi = 180^\circ\ (\max A_p)$$

$$\text{CB}:\ A_i = \alpha,\ \Delta\phi = 0^\circ \quad | \quad \text{CC}:\ A_v \approx +1$$

WHY / Emitter follower (CC) has voltage gain +1 (0-deg phase) and high input / low output impedance — ideal unity-gain buffer driving heavy low-resistance loads!

WATCH / Adding an unbypassed emitter resistor RE stabilizes Q-point thermally and sets rock-solid voltage gain Av ~ -RC / RE independent of transistor beta spread!

Transfer curve Vout(Vin) & CMOS/RTL NOT gate origin

Sweeping DC input Vin from 0 V to VCC sweeps Vout = VCE across three sharp zones that define BOTH linear amplification and digital inversion:

$$V_{\text{in}} < 0.6\ \text{V}\ (\text{Cutoff}):\ V_{\text{out}} = V_{CC}\ \ (\text{Logic } 0 \to \text{Logic } 1)$$

$$V_{\text{in}} > V_{\text{sat}}\ (\text{Saturation}):\ V_{\text{out}} = V_{CE,\text{sat}} \approx 0.2\ \text{V}\ \ (1 \to 0)$$

WHY / Analog amplifiers bias RIGHT IN THE MIDDLE steep linear slope; digital logic gates slam cleanly between the TWO FLAT ENDS (0 V and VCC) where transistor power is tiny!

WATCH / In active region P_transistor = VCE IC is maximum (hot!); in Cutoff (IC = 0) and hard Saturation (VCE ~ 0.2 V), switch dissipation I_C V_{CE} is nearly ZERO!

Designing a saturated BJT lamp/relay switch (E11 & P25)

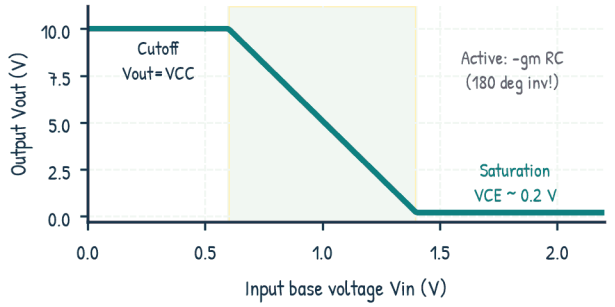
To switch load I_{C,sat} cleanly from logic high V_{in} (e.g. 5.0 V MCU pin), force base current safely above the minimum threshold I_{B,min} = I_{C,sat} / beta:

$$I_B = \frac{V_{\text{in}} - V_{BE,\text{on}}}{R_B} \geq \text{OD} \cdot \frac{I_{C,\text{sat}}}{\beta}, \qquad R_B \leq \frac{V_{\text{in}} - 0.7\ \text{V}}{\text{OD} \cdot (I_{C,\text{sat}}/\beta)}$$

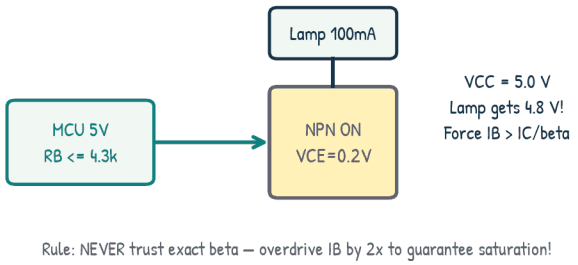
WHY / E11 check: 5 V supply, 100 mA lamp, beta = 100 needs IB >= 1.0 mA => RB <= (5 - 0.7)/1 mA = 4.3 kOhm (pick 2.2 kOhm for 2x overdrive); lamp gets 5.0 - 0.2 = 4.8 V!

WATCH / When driving inductive relay/motor coils with a BJT switch, ALWAYS place a flyback diode anti-parallel across the coil so L(dI/dt) turn-off spikes never puncture VCE!

Common-emitter voltage transfer curve Vout(Vin) and 180-deg phase inversion



Hard-saturated NPN lamp/relay switch with 2x base-current overdrive



Rule: NEVER trust exact beta — overdrive IB by 2x to guarantee saturation!

REMEMBER / CE amplifier inverts phase by 180 deg with Av = -beta RC / r_in = -gm RC and Ap = beta * |Av|; a digital BJT switch lives ONLY in Cutoff and Saturation!

Diode/transistor logic circuits, truth tables & waveforms

DL/RTL hardware implementations, all seven gate truth tables, and JEE pulse-timing traces.

Hardware Diode Logic (DL) & Transistor Inverter (RTL)

JEE Main regularly draws raw diode-resistor-transistor schematics and asks which Boolean gate they implement — identify the pull resistor FIRST:

$$\text{Diode OR}: \text{ anodes } A, B;\ \text{cathodes} + R_{\text{down}} \to 0$$

$$\text{Diode AND}: \text{ cathodes } A, B;\ \text{anodes} + R_{\text{up}} \to V_{CC}$$

WHY / In Diode AND, if EITHER input A = 0 V, its diode conducts and clamps output Y down to 0.7 V (Logic 0); Y stays HIGH (+VCC) ONLY when BOTH A = 1 AND B = 1!

WATCH / Two NPN transistors in PARALLEL sharing collector pull-up RC form a NOR gate; two NPN transistors in SERIES under pull-up RC form a NAND gate!

Complete truth tables of all 7 standard gates

For input pairs (A, B) ordered strictly as (0,0), (0,1), (1,0), (1,1), memorise each gate's 4-bit output signature cold:

$$\text{AND } (A \cdot B): \ 0,0,0,1 \qquad \text{NAND } (\overline{A \cdot B}): \ 1,1,1,0$$

$$\text{OR } (A + B): \ 0,1,1,1 \qquad \text{NOR } (\overline{A + B}): \ 1,0,0,0$$

$$\text{XOR } (A \oplus B): \ 0,1,1,0 \qquad \text{XNOR } (A \odot B): \ 1,0,0,1$$

WHY / Fast verbal shortcuts: AND = "all 1s"; OR = "any 1"; NAND = "0 ONLY at 1,1"; NOR = "1 ONLY at 0,0"; XOR = "inequality / odd-parity detector"; XNOR = "equality comparator"!

WATCH / Tying one input of XOR to a control line C makes a programmable inverter: A xor 0 = A (buffer), whereas A xor 1 = NOT A (inverter)!

Why binary logic beats decimal hardware (C14)

Digital circuits encode 0 and 1 as wide voltage bands separated by a forbidden transition zone, rejecting power noise, crosstalk, and device tolerances:

$$A \oplus B = A\overline{B} + \overline{A}B, \qquad A \odot B = \overline{A \oplus B} = AB + \overline{A}\,\overline{B}$$

WHY / Every CMOS logic gate regenerates degraded levels right back to clean 0 V or VDD rails — errors do NOT accumulate across 100 stages!

WATCH / Associativity superpower: multi-input XOR (A xor B xor C) outputs 1 whenever an ODD number of inputs equal 1 — instant parity checker!

Solving JEE digital pulse-train waveform problems

When given square-wave timing graphs A(t), B(t) and 4 candidate output graphs Y(t), NEVER trace intermediate gates sloppily by eye:

1. Slice time into clock slots $t_1, t_2, \ldots, t_k$
2. Evaluate $Y(A_i, B_i)$ slot by slot

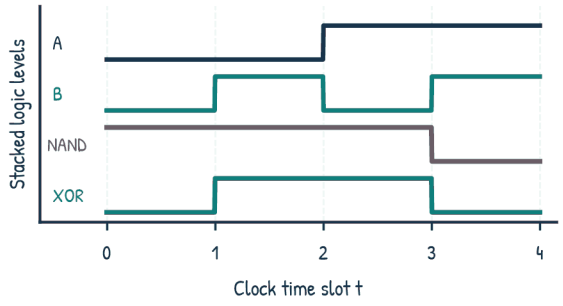
WHY / Simplifying the gate network algebraically FIRST (e.g. spotting Y = A + B' in 5 seconds!) eliminates 100% of intermediate-node copy errors under exam pressure!

WATCH / Watch whether HIGH voltage is drawn UP or whether negative logic is stated in the stem — read vertical axis labels before ticking options!

Diode OR/AND hardware realisations and six standard Boolean gate outputs



Digital pulse-train timing trace through NAND and XOR gates



REMEMBER / Diode anodes at inputs + pull-DOWN resistor = OR gate; diode cathodes at inputs + pull-UP resistor = AND gate; add a CE transistor stage to invert into NOR / NAND!

Boolean algebra, De Morgan, universal NAND/NOR & adders

Absorption laws, bubble-pushing De Morgan rules, minimum NAND/NOR counts, and half/full adders.

Boolean algebra shortcuts that collapse 5 lines to 1

Beyond commutativity and distributivity, three non-ordinary identities do 90% of all JEE Boolean simplifications:

$$A + A = A, \quad A + 1 = 1, \quad A + \overline{A} = 1, \quad A\overline{A} = 0$$

$$A + \overline{A}B = A + B, \quad (A + B)(A + C) = A + BC$$

WHY / Test A + A'B = A + B quickly: if A = 1, both equal 1; if A = 0, LHS = B and RHS = B — identical on EVERY row in 2 seconds!

WATCH / Never cancel terms like ordinary algebra (A + B = A + C does NOT imply B = C in Boolean algebra)! Always double-check any tricky reduction on the 4 truth rows!

De Morgan's theorems & graphical bubble-pushing

Break an overbar, flip the operator between AND (.) and OR (+)! Graphically: a NAND gate IS an inverted-input OR gate; a NOR gate IS an inverted-input AND gate:

$$\overline{A \cdot B} = \overline{A} + \overline{B}, \qquad \overline{A + B} = \overline{A} \cdot \overline{B}$$

WHY / Whenever two inversion bubbles meet head-to-head on the same wire, they CANCEL cleanly (double complement A'' = A) — instant visual reduction of multi-NAND schematics!

WATCH / Watch bar height carefully! OVERLINE(A + B) != OVERLINE(A) + OVERLINE(B)! Only split an overbar AT THE EXACT MOMENT you flip (+) <-> (.)!

Universal NAND & NOR minimum-gate counts

Both NAND and NOR are functionally complete (universal). Memorise BOTH construction formulas AND exact minimum 2-input gate counts:

$$\text{NAND}: \ [1, 2, 3, 4, 5] \ \ (\text{NOT}, \ \text{AND}, \ \text{OR}, \ \text{XOR}, \ \text{XNOR})$$

$$\text{NOR}: \ [1, 3, 2, 5, 4] \ \ (\text{NOT}, \ \text{AND}, \ \text{OR}, \ \text{XOR}, \ \text{XNOR})$$

WHY / Optimal 4-NAND XOR ingeniously reuses first shared gate G1 = NAND(A, B) to compute NAND(NAND(A, G1), NAND(B, G1)) = A xor B without two extra NOT gates!

WATCH / Classic JEE trap: naive student builds A'B + AB' with 5 NANDs and ticks "5" — MINIMUM number of 2-input NAND gates for XOR is ALWAYS 4!

Binary arithmetic: Half Adder & Full Adder (E12, Q28, P26)

Adding two 1-bit binary numbers A + B produces 2-bit output (Carry C, Sum S); adding carry-in C_in from a lower column makes a Full Adder:

$$\text{Half Adder}: \quad S = A \oplus B, \quad C = A \cdot B \quad (5 \text{ NAND})$$

$$\text{Full Adder}: \quad S = A \oplus B \oplus C_{\text{in}}, \quad C_{\text{out}} = AB + C_{\text{in}}(A \oplus B)$$

WHY / Check 1 + 1 in Half Adder: S = 1 xor 1 = 0, C = 1 . 1 = 1 -> binary "10" (= decimal 2)! Two Half Adders + one OR gate build a complete 1-bit Full Adder (9 NANDs min)!

WATCH / Half subtractor calculating A - B has the EXACT SAME Difference bit D = A xor B, changing ONLY Borrow = A' . B!

Universal gate synthesis table: exact minimum NAND and NOR gate counts

NOT(A) = NAND(A, A) -> 1 NAND | 1 NOR

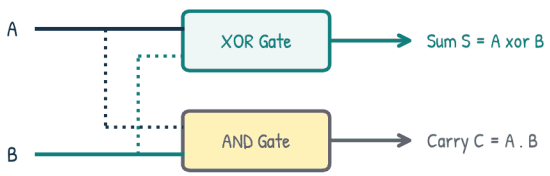
AND(A, B) = NOT(NAND(A,B)) -> 2 NAND | 3 NOR

OR(A, B) = NAND(A', B') -> 3 NAND | 2 NOR

XOR(A, B) = optimal 2-level -> 4 NAND | 5 NOR

XNOR(A,B) = dual symmetry -> 5 NAND | 4 NOR

Half adder combinational circuit generating Sum (XOR) and Carry (AND)



1 + 1 = 10_2 (C=1, S=0); Full Adder = 2 Half Adders + 1 OR gate!

REMEMBER / Minimum gate count sequence: NAND builds [NOT, AND, OR, XOR, XNOR] in [1, 2, 3, 4, 5] gates; NOR reverses AND/OR & XOR/XNOR to [1, 3, 2, 5, 4] gates!

Master JEE triage, multi-diode networks & 10 exam traps

Rapid formula selector, multi-diode open-circuit solver, LED lookup, and ten fatal traps.

Multi-diode circuit solver (Part 7.3 gold standard)

When a network contains two or more diodes sharing a node (e.g. diodes D1 at +1 V and D2 at +3 V feeding one pull-down resistor), DO NOT guess randomly:

1. Open diodes → rank open circuit $V_A - V_K$
2. Turn ON largest $(V_A - V_K > V_\gamma)$, re solve!

WHY / Example: anodes at +3 V and +1 V joined to ground via R (ideal diodes) -> +3 V diode turns ON first, clamping cathode node to +3 V so the +1 V diode is REVERSE biased at -2 V (OFF)!

WATCH / NEVER leave a solved circuit with an "ON" diode carrying negative current I < 0 or an "OFF" diode sitting at forward V > V_knee!

Core silicon 300 K anchor constants (memorise cold!)

Carry these eight room-temperature numbers into every JEE Main, JEE Advanced, and NSEP paper for instant 5-second sanity checks:

$V_T = 25.9\ \text{mV}, \quad \Delta V_{10\times} = 60\ \text{mV/dec}, \quad hc = 1240\ \text{eV}\cdot\text{nm}$

$n_i = 1.5 \times 10^{16}\ \text{m}^{-3}, \quad \rho_i \approx 2300\ \Omega\cdot\text{m}$

$V_\gamma \approx 0.7\ \text{V (Si)}, \qquad V_{CE,\text{sat}} \approx 0.2\ \text{V}$

WHY / Even when a problem gives slightly rounded data (VT = 25 mV or ni = 10^16 m^-3), knowing exact physical scales catches 10^6 unit slips instantly!

WATCH / Germanium numbers differ noticeably: Eg = 0.66 eV, V_knee ~ 0.3 V, V_{CE,sat} ~ 0.1 V, and much larger leakage current I0 (uA vs nA)!

Five conceptual JEE traps examiners repeat every year

1. "n-type silicon is negatively charged" -> FALSE! Mobile e- + fixed P^+ sum to ZERO net charge!
2. "Depletion region carries net charge" -> FALSE! Equal +e Nd xn and -e Na xp dipole!
3. "Voltmeter across p-n diode reads 0.7 V" -> FALSE! Contact potentials cancel to 0 V!

$\text{LED brightness} \propto I_F \quad (E_g \text{ sets ONLY } \lambda!)$

WHY / Raising forward LED current emits MORE photons/sec of the SAME wavelength 1240/Eg; changing colour requires changing semiconductor alloy composition x!

WATCH / 5. "Transistor amplifies energy" -> FALSE! DC power supply VCC provides the energy; transistor merely valves collector flow!

Five circuit-algebra JEE traps to check before submitting

Run this 30-second checklist across your numerical working before locking options:

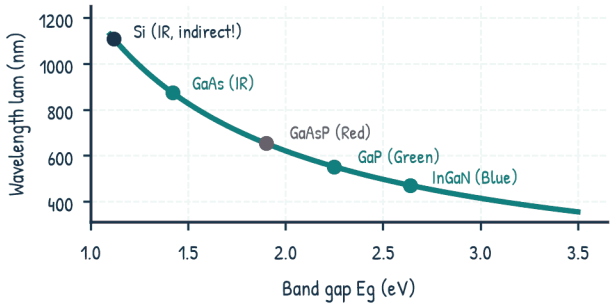
$\text{Bridge drop} = 2V_\gamma \mid \Delta V_{\text{FW}} = \frac{I}{2fC} \text{ vs } \Delta V_{\text{HW}} = \frac{I}{fC}$

$\text{Check } V_{\text{th}} > V_Z, \quad P_{Z,\max}\big|_{I_L=0}, \quad \beta I_B \text{ vs } I_{C,\text{sat}}$

WHY / Nearly 80% of all lost marks in JEE electronics come NOT from hard physics, but from applying IC = beta IB in saturation or forgetting 2f vs f in ripple!

WATCH / Always check cm^(-3) vs m^(-3) BEFORE using sigma = n e mu or n p = ni^2 — missing 10^6 gives answers off by 1,000,000x!

Direct-gap photon wavelength lam = 1240 / Eg across LED materials



Systematic open-circuit ranking protocol for multi-diode networks

4-Step Multi-Diode Circuit Protocol

1. Open all diodes -> rank open-circuit anode-cathode voltages V_open

2. Turn ON diode with LARGEST positive V_open first (0.7 V source)

3. Re-compute node voltages -> check whether 2nd diode also exceeds 0.7 V

4. Verify EVERY ON diode has I > 0 and EVERY OFF diode has V < V_knee!

REMEMBER / Mode BEFORE algebra on every diode, Zener, and transistor problem — one 5-second open-circuit / saturation check saves full marks!

Scaled Bohr donors, Fermi level & intrinsic crossover

Hydrogenic impurity derivation, valley-anisotropy honesty factor, Fermi shifts, and 560 K ceiling.

Hydrogenic impurity model (OL1, E4, P22 & P28)

An extra fifth valence electron of P orbiting a +e core inside crystal permittivity kappa = eps_r moves with band effective mass m*, rescaling Bohr's atom cleanly:

$$E_d = 13.6\ \text{eV}\,\frac{m^*/m_e}{\kappa^2}, \qquad a^* = a_0\,\frac{\kappa}{m^*/m_e} \quad (a_0 = 0.0529\ \text{nm})$$

$$\text{Si}\ (\kappa = 11.7,\ 0.26\,m_e):\ E_d = 26\ \text{meV},\ a^* = 2.4\ \text{nm}$$

$$\text{GaAs}\ (\kappa = 12.9,\ 0.067\,m_e):\ E_d = 5.5\ \text{meV},\ a^* = 10.2\ \text{nm}$$

WHY / Because a* = 2.4 nm spans ~4.4 cubic lattice cells (a_Si = 0.543 nm; ~9 bond lengths!), averaging over hundreds of Si atoms makes macroscopic dielectric screening kappa valid!

WATCH / Olympiad marking-rubric MUST-WRITE honesty line (Part 12): measured Si:P is 45 meV (1.7x higher) because silicon's 6 anisotropic conduction valleys & central-cell potential deviate from isotropic hydrogen!

Fermi-Dirac distribution & Fermi-level shift with doping

Occupation probability of state E is governed by Fermi-Dirac statistics; when E - EF >> kT, it reduces to Boltzmann factor exp(-(E - EF)/kT):

$$f(E) = \frac{1}{e^{(E-E_F)/(k_BT)}+1}, \qquad E_{F,i} = \frac{E_c+E_v}{2} + \frac{3}{4}k_BT\ln\left(\frac{m_h^*}{m_e^*}\right)$$

$$n = n_i\,e^{(E_F-E_i)/(k_BT)}, \qquad p = n_i\,e^{(E_i-E_F)/(k_BT)}$$

WHY / Multiplying n and p cancels EF identically to prove np = ni^2! n-doping shifts EF UP toward Ec by kT ln(Nd/ni); p-doping shifts EF DOWN toward Ev by kT ln(Na/ni)!

WATCH / At T -> 0 K in pure crystal, intrinsic Fermi level sits EXACTLY midway at (Ec + Ev)/2 regardless of mass asymmetry m_h* != m_e*!

Freeze-out (<120 K), extrinsic plateau & intrinsic ceiling

Plotting log n vs 1000/T across cryogenic to high temperature reveals three distinct physical regimes in doped silicon:

1. Freeze out $(k_BT \ll E_d):\ n \propto e^{-E_d/(2k_BT)}$
2. Extrinsic saturation (120 to 500 K): $n \approx N_d = \text{const!}$
3. Intrinsic $(n_i > N_d):\ n \approx n_i \propto T^{3/2}e^{-E_g/(2k_BT)}$

WHY / Notice factor 2 in BOTH slopes on an Arrhenius plot: steep high-T slope measures -Eg/(2k), shallow cryogenic slope measures -Ed/(2k)!

WATCH / In the extrinsic saturation window (near 300 K), n = Nd is constant while mu ~ T^(-3/2) falls — so lightly doped Si actually shows a SMALL POSITIVE dR/dT until ni wakes up!

When silicon devices forget their doping (OL12 exact)

Once thermal generation raises ni(T) up to dopant density Nd, p-n junctions lose rectification (V0 -> 0) and transistors fail completely:

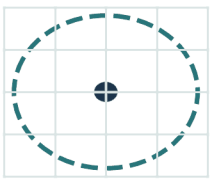
$$\left(\frac{T}{300}\right)^{3/2}\exp\left[\frac{E_g}{2k_B}\left(\frac{1}{300}-\frac{1}{T}\right)\right] = \frac{N_d}{n_i(300\ \text{K})} = 6.67\times10^4$$

$$\text{Si}\ (N_d = 10^{21}\ \text{m}^{-3}):\ T_{\max} \approx 560\ \text{K}\ (\approx 287^\circ\text{C})$$

WHY / Wide-bandgap semiconductors SiC (Eg = 3.26 eV) and GaN (Eg = 3.4 eV) push this thermal ceiling well past 600 deg C — why EV inverters & jet-engine sensors use SiC/GaN!

WATCH / Germanium (Eg = 0.66 eV) hits intrinsic washout right around ~90-100 deg C — why early 1950s Ge transistors failed in hot summer equipment and Si won!

Dielectric-screened effective-mass Bohr orbit of phosphorus in silicon



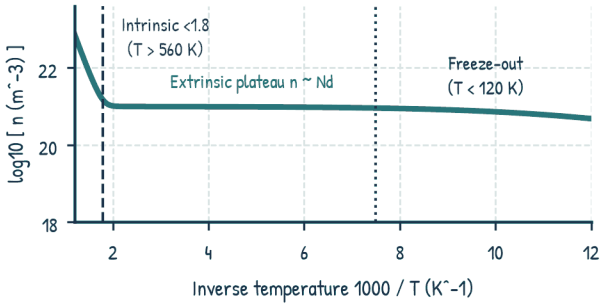
a* = 2.4 nm (~4.4 Si cells)

Scaled Hydrogenic Model

Ed = 13.6 (m*/m) / kappa^2
Si:P calc = 26 meV ~ kT
Measured = 45 meV (x1.7)

GaAs: Ed = 5.5 meV,
a* = 10.2 nm (shallow!)

Three temperature regimes of doped silicon: freeze-out, extrinsic, intrinsic



REMEMBER / Scaled Bohr gives Ed = 13.6 (m*/m) / kappa^2 = 26 meV ~ kT and a* = 2.4 nm in Si (measured 45 meV due to 6-valley anisotropy); Si loses doping above ~560 K!

Boltzmann diode derivation, Poisson step junction & varactors

Minority injection law, 2-sided vs 1-sided depletion audit, and voltage-tuned RF varactors.

Shockley equation from Boltzmann boundary law (OL2)

In equilibrium, minority electrons at the p-side depletion edge equal n_{p0} = n_{n0} exp(-eV0/kT). Forward bias V lowers barrier to V0 - V, boosting edge density exponentially:

$$n_p(-x_p) = n_{p0}\, e^{V/V_T} \Rightarrow \Delta n_p = n_{p0}(e^{V/V_T} - 1)$$

$$I = I_0(e^{V/V_T} - 1), \qquad I_0 = eA\left(\frac{D_p\, p_{n0}}{L_p} + \frac{D_n\, n_{p0}}{L_n}\right)$$

WHY / Using p_{n0} = ni^2 / Nd and n_{p0} = ni^2 / Na immediately proves WHY reverse leakage I0 scales as ni^2 and why heavily doping one side suppresses its reverse injection!

WATCH / Diffusion length L_n = sqrt(Dn tau_n) is how far injected minority electrons travel before recombining; a diode needs base thicker than L_n, whereas a BJT needs W_B << L_n!

Exact step-junction Poisson audit: 2-sided vs 1-sided (OL3/P29)

Audit distinction (OL3 & P29): total depletion width W = xp + xn across acceptor Na and donor Nd sides depends on REDUCED doping density N_eff:

$$W = \sqrt{\frac{2\varepsilon(V_0 + V_R)}{e\, N_{\text{eff}}}}, \qquad N_{\text{eff}} = \frac{N_a N_d}{N_a + N_d}$$

$$W_{1-\text{side}} = \sqrt{\frac{2\varepsilon(V_0 + V_R)}{eN}}, \qquad W_{\text{sym}} = \sqrt{2}\, W_{1-\text{side}}$$

WHY / In P29 (eps = 11.7 eps0, N = 10^21 m^-3, V0 + VR = 5.0 V, A = 1 mm^2): p+n one-sided gives W = 2.55 um, Cj = 40.7 pF; symmetric Na = Nd = N gives W = 3.60 um, Cj = 28.8 pF!

WATCH / Always state explicitly in Olympiad solutions whether you model a one-sided p^+n junction (N_eff = N) or symmetric junction (N_eff = N/2) — instant full rigor credit!

Varactor (variable-reactance) tuning diodes & 1/C^2 law

Small AC signal wiggle dV around reverse bias VR charges/discharges depletion edges separated by W(VR) exactly like a voltage-controlled parallel-plate capacitor:

$$C_j(V_R) = \frac{dQ}{dV_R} = \frac{\varepsilon A}{W(V_R)} = \frac{C_{j0}}{\sqrt{1 + V_R/V_0}}$$

$$\frac{1}{C_j^2} = \frac{2(V_0 + V_R)}{e\,\varepsilon\, A^2 N_{\text{eff}}} \quad (\text{STRICT STRAIGHT LINE vs } V_R!)$$

WHY / Plotting 1/Cj^2 against reverse bias VR in the lab measures BOTH built-in potential V0 (horizontal intercept -V0!) AND substrate doping N_eff (from slope)!

WATCH / Putting a varactor inside an LC tank f0 = 1 / (2 pi sqrt(L Cj)) gives f0 ~ (V0 + VR)^(+1/4) — how every PLL, FM modulator, and radio tuner tunes frequency electronically!

Linearly graded junctions & forward diffusion capacitance

Not every junction is a step function! Diffused junctions with rho(x) = e a x obey a cube-root law, while forward-biased diodes add a huge storage capacitance C_diff:

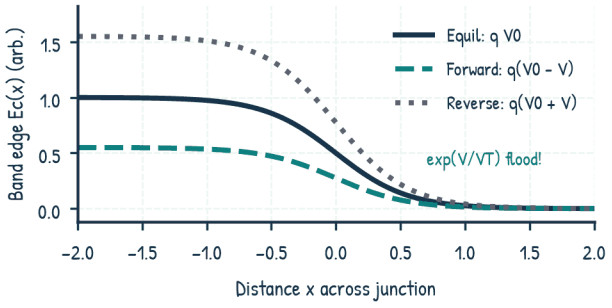
$$\text{Graded } \rho = eax: \quad W \propto (V_0 + V_R)^{1/3} \Rightarrow C_j \propto (V_0 + V_R)^{-1/3}$$

$$\text{Forward storage}: \quad Q_{\min} = \tau_T I \Rightarrow C_{\text{diff}} = \frac{dQ_{\min}}{dV} = \frac{\tau_T I}{V_T}$$

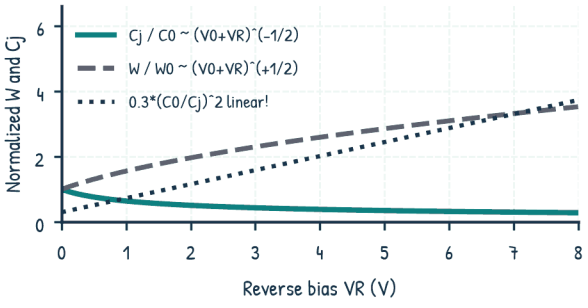
WHY / Stored minority charge Q_min = tau_T I takes transit/lifetime time tau_T to clear on sudden turn-off — why ordinary 1N4007 diodes fail at 100 kHz and Schottky diodes (majority carriers, tau_T ~ 0!) rule SMPS!

WATCH / Hyperabrupt varactors engineer doping N(x) ~ x^(-3/2) so Cj ~ (V0 + VR)^(-2), making LC resonant frequency f0 ~ 1/sqrt(Cj) STRICTLY LINEAR in control voltage VR!

Bias modulation of conduction-band barrier hill q(V0 - V) across a junction



Depletion width W(VR), varactor Cj(VR), and linear 1/Cj^2 extraction



REMEMBER / Watch 1-sided vs symmetric step junctions: W_1side = sqrt(2 eps V_tot / eN), whereas symmetric Na = Nd = N has W_sym = sqrt(4 eps V_tot / eN) = sqrt(2) W_1side!

Hall effect metrology, diode thermometry & 1.22 V reference

Lorentz carrier-sign determination, two-carrier Hall zero, -2 mV/K sensor, and Brokaw bandgap circuit.

Hall effect derivation, polarity & carrier extraction (OL6)

Pass current I_x along +x through rectangular slab (thickness t along B_z). Magnetic force q v_d x B pushes BOTH drift electrons (v_x < 0, q = -e) and drift holes (v_x > 0, q = +e) toward the SAME face!

$$|E_y| = v_d B_z = \frac{J_x B_z}{ne} \Rightarrow V_H = \frac{IB}{net}, \qquad R_H \equiv \frac{E_y}{J_x B_z} = \mp\frac{1}{ne}$$

WHY / OL6 & P33 check: I = 10 mA, B = 0.5 T, t = 0.5 mm, V_H = 62 mV => n = (0.01 x 0.5)/(1.6e-19 x 0.062 x 5e-4) = 1.01 x 10^21 m^(-3); doubling V_H halves n (P20: 31 mV -> 2.0 x 10^21 m^-3)!

WATCH / Notice slab WIDTH w cancels out of V_H = I B / (n e t)! Thin epitaxial films (tiny t) generate huge, easily measured millivolt Hall signals!

Combining Hall + conductivity & two-carrier Hall zero

Conductivity alone gives ONLY the product n mu; pairing a 4-probe resistivity test with a Hall measurement extracts carrier density n AND mobility mu SEPARATELY:

$$|R_H| = \frac{1}{ne}, \qquad \mu_H = |R_H|\,\sigma = \frac{V_H L}{W B V_{\text{long}}}$$

$$R_{H,\,\text{ambipolar}} = \frac{p\mu_h^2 - n\mu_e^2}{e(p\mu_h + n\mu_e)^2} \quad (\text{vanishes when } p\mu_h^2 = n\mu_e^2!)$$

WHY / Look at mu^2 in the numerator of R_H! Because mu_e = 2.81 mu_h in Si, intrinsic Si (n = p = ni) actually shows an N-TYPE negative Hall coefficient!

WATCH / Classic IPhO trap: in lightly p-doped silicon where p = (mu_e/mu_h)^2 n ~ 7.9 n, Hall voltage passes right through ZERO even though mobile carriers are abundant!

Diode & BJT thermometer: dV_BE/dT = -2 mV/K (OL5)

Driving a diode-connected BJT with constant current source I_C = const and differentiating V_BE(T) = V_T ln(I_C / I_0(T)) with I_0 ~ T^3 exp(-Eg / kT):

$$\frac{dV_{BE}}{dT} \approx \frac{V_{BE} - (E_g/e + 3V_T)}{T} \approx \frac{0.70 - 1.20}{300\ \text{K}} \approx -1.7 \text{ to } -2.1\ \text{mV/K}$$

WHY / Even though V_T = kT/e tries to raise V_BE, saturation current I_0(T) explodes exponentially across Eg, pulling V_BE DOWN linearly across -55 to +150 deg C (CTAT = Complementary To Absolute T)!

WATCH / Every CPU/GPU die embeds on-chip BJT thermal diodes reading this exact -2 mV/K slope to throttle clock frequency before silicon overheats!

Bandgap voltage reference: cancelling CTAT + PTAT = 1.22 V

Run two matched BJTs at fixed current-density ratio J1 / J2 = M (e.g. M = 10): their V_BE difference eliminates I_0 completely to create a pure +T signal (PTAT):

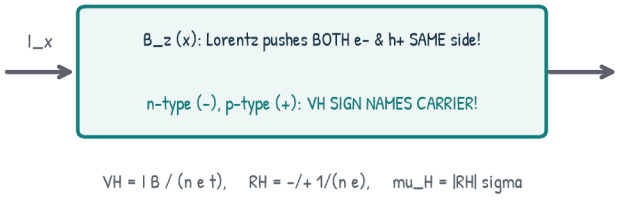
$$\Delta V_{BE} = V_{BE1} - V_{BE2} = V_T \ln M = \frac{k_B \ln M}{e} T \quad (\text{pure PTAT!})$$

$$V_{\text{ref}} = V_{BE} + K\Delta V_{BE} \Rightarrow \frac{dV_{\text{ref}}}{dT} = 0 \quad (1.22\ \text{V})$$

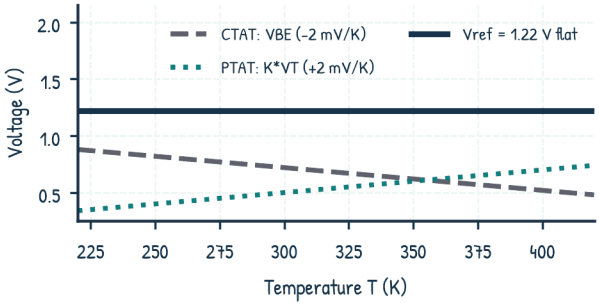
WHY / Summing falling V_BE (-2 mV/K, extrapolating to Eg/e at 0 K) + amplified rising K*dV_BE (+2 mV/K) locks V_ref right onto silicon's extrapolated 0 K bandgap ~1.22-1.25 V (P34)!

WATCH / Every precision IC regulator (LM317's famous 1.25 V reference, ADC refs, phone PMICs) rests directly on this CTAT + PTAT cancellation!

Hall effect slab geometry: Lorentz deflection names carrier sign and density



Cancelling CTAT (-2 mV/K) and PTAT (+2 mV/K) slopes in a 1.22 V reference



REMEMBER / Lorentz force deflects e- and h+ to the EXACT SAME transverse face — so Hall voltage polarity unambiguously distinguishes n-type from p-type!

Shockley-Queisser PV bound, shaded strings & tandem cells

Spectral loss trade-off (~33.7% at 1.34 eV), tandem solar stacks, hotspot physics, and bypass diodes.

Two fundamental spectral losses in single-junction PV (OL4)

Sunlight spans ~0.4 to 4.0 eV (~5800 K blackbody). A single semiconductor bandgap Eg extracts EXACTLY Eg (minus recombination loss) per absorbed photon:

$$h\nu < E_g: \text{ unabsorbed transmission loss } (0\%)$$

$$h\nu > E_g: \text{ excess } (h\nu - E_g) \text{ lost as phonons}$$

WHY / Tiny Eg absorbs everything but delivers tiny V_oc < Eg/e; huge Eg gives large V_oc but wastes nearly all solar photons -> product peaks sharply around Eg ~ 1.34 eV!

WATCH / In P30 Olympiad answers, you MUST name BOTH spectral losses PLUS radiative recombination (which forces V_oc < Eg/e and FF < 1) to derive the ~33.7% ceiling!

Why crystalline Si (1.12 eV) dominates & how tandems beat 33%

Silicon's 1.12 eV gap sits right beside the broad SQ plateau (~32% theoretical limit; ~27% lab record), backed by abundant SiO2 sand and wafer purity.

$$\text{Tandem}: \quad E_{g,\,\text{top}} \approx 1.7 \text{ eV}, \quad E_{g,\,\text{bot}} = 1.12 \text{ eV}$$

$$\eta_{\max}: \ 33.7\% \ (1\text{J}) \rightarrow 45\% \ (2\text{J}) \rightarrow 68\% \ (\infty\text{J})$$

WHY / Top wide-gap subcell harvests blue/green photons at HIGH voltage (~1.2 V) while letting red/IR pass transparently down into bottom Si subcell — crushing thermalisation loss!

WATCH / Focusing sunlight with lenses/mirrors (concentrator PV, C suns) boosts I_L -> raises V_oc by V_T ln C, lifting theoretical infinite-junction limit toward 86.8%!

Shaded series solar cell hotspot crisis (OL10 & P35)

Solar modules wire N cells (e.g. 60 or 72) IN SERIES so voltages add to ~35-40 V. If ONE single leaf shades cell #k, KCL forces identical current through every cell:

$$I_{\text{str}} > I_{L,\,\text{sh}} \Rightarrow V_{\text{shaded}} = V_{\text{load}} - \sum_{j \neq k} V_j < 0$$

$$P_{\text{hotspot}} = |V_{\text{shaded}}|\, I_{\text{string}} \quad \text{(tens of watts in ONE cell!)}$$

WHY / All N - 1 sunlit cells gang up like a 30 V battery ramming 8 A BACKWARD into the one shaded cell — baking EVA encapsulant and cracking glass!

WATCH / Without protection, either open-circuit load drops string power to the worst cell OR low load resistance drives destructive reverse avalanche breakdown!

Antiparallel bypass diodes & stepped multi-peak I-V curves

Wiring a Schottky bypass diode in ANTI-PARALLEL across each sub-string of M cells (typically 20-24 cells per diode, 3 diodes per 60/72-cell panel) saves both power and hardware:

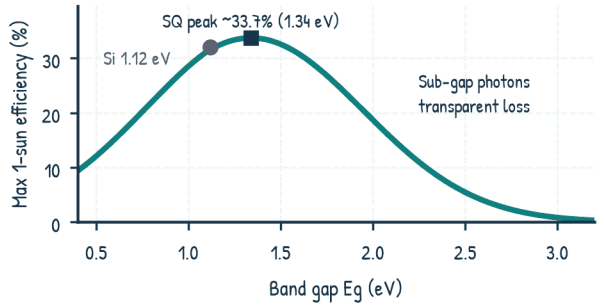
$$\text{Full sun } (V_{\text{blk}} > 0): \text{ bypass diode OFF}$$

$$\text{Shaded } (V_{\text{blk}} \approx -0.4 \text{ V}): \text{ bypass diode ON!}$$

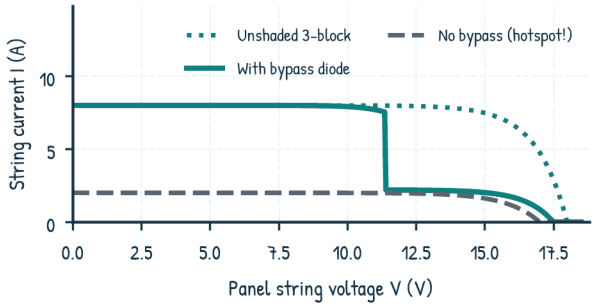
WHY / Full 8 A string current detours cleanly around the shaded sub-string: the module loses ONLY 1/3 of voltage (~65% power saved!) instead of collapsing by 80-90%!

WATCH / Signature on panel I(V) curve (P35c): K distinct current steps / local power peaks — which is why solar inverters run global MPPT sweeps!

Single-junction Shockley-Queisser limit (~33.7% near Eg = 1.34 eV)



Series PV string current collapse under shading vs bypass-diode step recovery



REMEMBER / Single-junction PV cells lose sub-gap red photons (unabsorbed) and excess above-gap blue energy (phonons), capping 1-sun efficiency near ~33.7% at Eg ~ 1.34 eV!

Shot noise, Johnson noise, NEP & LED load-line stability

Poisson discrete-charge floor, resistor thermal noise, minimum detectable power, and OL9 design.

Quantum shot noise from Poisson arrival statistics (OL8, P32)

Current I consists of discrete independent electrons e crossing a potential barrier at mean rate N / tau = I / e. Over observation window tau = 1 / (2 df), count variance is N:

$$\sigma_I^2 = (\tfrac{e}{\tau})^2 N = \tfrac{e^2}{\tau^2}(\tfrac{I\tau}{e}) = 2eI\,\Delta f \Rightarrow i_{\text{rms}} = \sqrt{2eI\,\Delta f}$$

WHY / OL8 & P32 checks: (1) I = 1 uA, df = 1 MHz => i_rms = 0.566 nA; (2) I = 4 uA, df = 2 MHz => i_rms = 1.60 nA! Narrowing receiver bandwidth df by 4x cuts rms noise in HALF!

WATCH / Dark current I_d and background light I_bg add uncorrelated shot noise even when signal modulates around zero: total shot variance is 2 e (I_sig + I_dark + I_bg) df!

Johnson-Nyquist resistor noise & photoreceiver SNR / NEP

Load resistor R_L at temperature T adds thermal Johnson current noise independent of DC current; total receiver noise sums mean-square powers:

$$i_{n,\,\text{tot}} = \sqrt{(2e(I_{\text{photo}} + I_{\text{dark}}) + \tfrac{4k_BT}{R_L})\Delta f}$$

$$\text{SNR} = \frac{I_{\text{photo}}^2}{i_{n,\,\text{tot}}^2}, \qquad \text{NEP} = \frac{i_{n,\,\text{tot}}|_{\Delta f = 1\ \text{Hz}}}{\mathcal{R}_\lambda} \quad (\text{W}/\sqrt{\text{Hz}})$$

WHY / Shot noise beats thermal noise once I*R_L > 2 k_B T / e = 51.7 mV! Any voltage drop > 52 mV across load resistor R_L puts the receiver in the quantum shot-noise regime!

WATCH / Never add rms currents linearly (i1 + i2)! Independent Gaussian/Poisson noise sources ALWAYS add IN QUADRATURE: sqrt(i_shot^2 + i_thermal^2)!

Why LEDs MUST have current-limiting ballast (OL9)

An LED is an exponential diode (I ~ exp(V / VT)), NOT an ohmic bulb! Connecting a 2.0 V LED directly across a stiff voltage rail invites instant thermal runaway:

$$R_{\text{series}} = \frac{V_{CC} - V_{\text{LED}}}{I_{\text{target}}} \quad (\text{OL9}: \ \frac{5.0 - 2.0\ \text{V}}{20\ \text{mA}} = 150\ \Omega, \ P_R = 60\ \text{mW})$$

WHY / Without R_series, a +100 mV supply bump or +50 K junction warm-up (-100 mV knee shift!) boosts LED current by 10^(100/60) ~ 46x — melting the bond wire in milliseconds!

WATCH / Never wire two bare LEDs directly IN PARALLEL off one shared resistor! The slightly warmer LED drops V_LED by 2 mV/K, hogs all current, overheats further, and cascades to destruction!

Small-signal sensitivity: voltage source vs series resistor

Compare fractional current sensitivity dI / I to a diode knee shift dV_LED under stiff voltage drive vs resistor-ballasted drive:

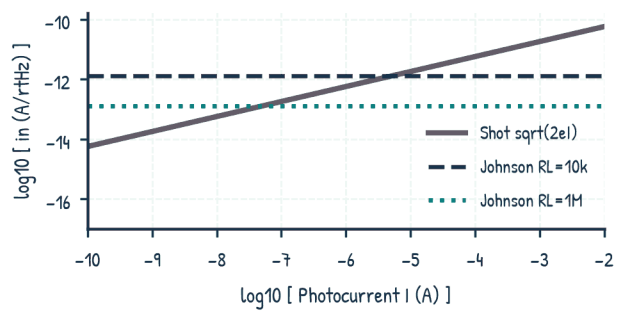
$$\text{Stiff V drive}: \ \frac{\delta I}{I} = -\frac{\delta V_{\text{LED}}}{V_T} \approx -39 \times \delta V_{\text{LED}} \ (\text{in volts!})$$

$$\text{With } R_s: \ \frac{\delta I}{I} = -\frac{\delta V_{\text{LED}}}{IR_s + V_T} \approx -\frac{\delta V_{\text{LED}}}{V_{CC} - V_{\text{LED}}} \ (115\text{x gentler!})$$

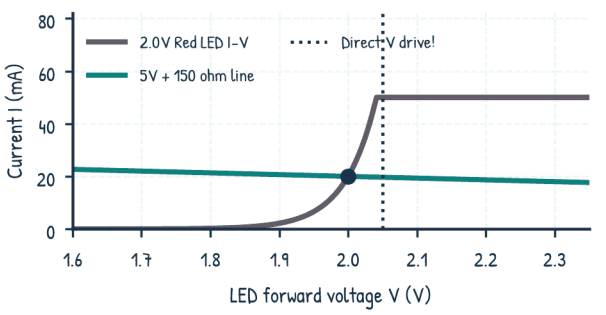
WHY / Dropping 3.0 V across R_s makes loop slope resistance I R_s + V_T = 3.026 V instead of 0.026 V — turning a wild exponential device into a rock-steady 3% current source!

WATCH / High-power white LED headlamps replace wasteful series resistors with buck switch-mode constant-current drivers (>92% efficiency vs 40% linear resistor)!

Quantum shot noise floor sqrt(2eI) vs Johnson-Nyquist resistor noise



Why LEDs need a series resistor: linear load line vs exponential runaway



REMEMBER / Shot noise i_rms = sqrt(2 e I df) comes from discrete charge quanta e; Johnson noise i_th = sqrt(4 kT df / R) comes from thermal agitation — add variances i_tot^2 = i_shot^2 + i_th^2!

Moore's quantum wall, oxide tunnelling & chip heat budget

Heisenberg confinement spread, WKB gate leakage, high-k FinFETs, and 500–800 W CMOS power.

Quantum wall #1: Heisenberg confinement energy (OL7, P31)

Squeezing channel/well dimension dx to a few nanometres gives carriers minimum zero-point kinetic energy spread via dx * dp >= hbar:

$$\Delta E \sim \frac{\hbar^2}{2m_e(\Delta x)^2}: \ 1.5 \text{ meV } (5 \text{ nm}), \ 4.2 \text{ meV } (3 \text{ nm})$$

$$\text{With } m^* = 0.19\, m_e: \ \Delta E(3 \text{ nm}) \approx 22 \text{ meV} \sim k_B T$$

WHY / Carrier de Broglie wavelength at 300 K is h / sqrt(3 m* kT) ~ 12 nm — smaller than a modern transistor pitch, so 2D/1D quantum subbands replace bulk classical drift!

WATCH / Quote BOTH free-electron mass result (4.2 meV at 3 nm, P31a exact!) AND note that light effective mass m* raises confinement right up to thermal kT = 26 meV!

Quantum wall #2: WKB gate-oxide tunnelling leakage

Scaling planar MOSFET capacitance C_ox = eps_ox A / d_ox once forced SiO2 thickness down to d_ox ~ 1.2 nm (~5 atomic layers!), where WKB tunnelling explodes:

$$T_{\text{WKB}} \sim \exp\left(-2\, d_{\text{ox}} \frac{\sqrt{2m_{\text{ox}}^* \Phi_B}}{\hbar}\right) \sim e^{-11.4\, d_{\text{ox}}(\text{nm})}$$

WHY / Every 0.2 nm thinning multiplies gate leakage current by ~10x! At 1.2 nm, 10^9 "OFF" transistors leak tens of amps straight through the gate insulator!

WATCH / How did semiconductor physics beat both walls (P31c)? (1) High-k HfO2 (kappa ~ 25 vs 3.9 -> 6x thicker physical wall for same C_ox!) + (2) 3D FinFET / Gate-All-Around nanosheets!

CMOS capacitor charging energetics & cycle audit (OL11, P36)

Reconciling single-transition vs full-cycle formulas cleanly from exact capacitor physics:

$$\text{Per edge } (0 \to 1 \text{ or } 1 \to 0): \ E_{\text{diss}} = \tfrac{1}{2} C V_{DD}^2$$

$$\text{Full cycle } (0 \to 1 \to 0): \ P_{\text{dyn}} = \alpha_{\text{sw}} N C V_{DD}^2 f$$

WHY / Audit clarity: if N gates each make 1 transition per clock period (activity alpha_sw = 0.5), P = (1/2) N C V^2 f = 500 W (OL11: N=10^9, f=1 GHz, C=1 fF, V=1 V) & 810 W (P36)!

WATCH / Making transistor channel R -> 0 does NOT reduce switching energy 1/2 CV^2 — it only burns the exact same 1/2 CV^2 faster in shorter RC time!

Why CPU clocks stalled near ~4–5 GHz (Dennard breakdown)

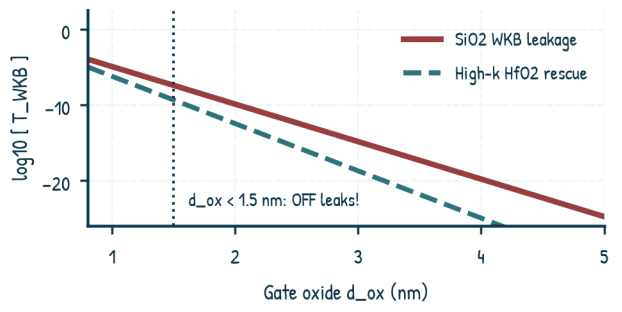
Total chip thermal dissipation combines dynamic switching power plus subthreshold leakage across billions of transistors:

$$P_{\text{total}} = \alpha_{\text{sw}} N C V_{DD}^2 f + N V_{DD} I_0\, e^{-V_{\text{th}}/(\eta V_T)}$$

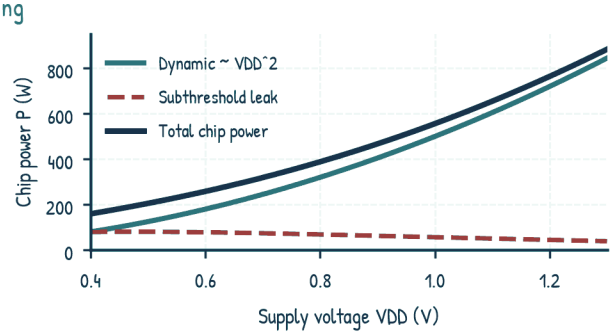
WHY / Because P_dyn scales quadratically with V_DD^2, dropping V_DD from 5 V -> 0.8 V cut switching energy by 39x! Below ~0.7 V, lowering threshold V_th makes exponential subthreshold leakage explode (60 mV/dec floor!) — freezing single-core GHz ~2005 and birthing multi-core CPUs!

WATCH / Notice how the ENTIRE chapter closes one circle: the 60 mV/decade Boltzmann limit of Sheet 05 IS the subthreshold leakage wall of 2026 2-nm chips!

WKB gate-oxide tunnelling probability and high-k HfO2 scaling rescue



Dynamic CV^2f vs subthreshold leakage trade-off across CMOS supply scaling



REMEMBER / Charging capacitor C through resistor to VDD draws CV^2 from battery: 1/2 CV^2 stored, 1/2 CV^2 burned on rise + 1/2 CV^2 burned on discharge = CV^2 per full 0->1->0 cycle!